

# ELI — Voice Cloning Pipeline

ELI v2.2.5

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## Voice Creation / Cloning Engine — end-to-end (2026-07-28)

How ELI turns a short audio/video clip of *any* voice — a character, an actor, a friend — into a usable ELI voice. Covers the full path from a dropped-in file to spoken output, the two distinct technologies involved, the dependency stack, and the real defects found and fixed while verifying it end-to-end on this machine.

### The two voice technologies (don't confuse them)

ELI actually ships **two** different ways to get a “character” voice, and they trade off very differently:

|                                                        | Character presets (voice_fx.py)                                     | Voice cloning (tts_xtts.py)                                  |
|--------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------|
| Built-in                                               | HAL, TARS, Rick, GLaDOS, JARVIS                                     | none — always from <i>your</i> clip                          |
| How it works                                           | base Piper voice + an ffmpeg -af effect chain (pitch/speed/filters) | Coqui XTTS-v2 zero-shot cloning from a ~6-20s reference clip |
| Fidelity                                               | an approximation / vibe                                             | reproduces the actual voice characteristics                  |
| Cost                                                   | free, instant, always available                                     | opt-in ~1.8GB neural model, GPU preferred                    |
| Covers characters with no preset (Cartman, Vader, ...) | no                                                                  | yes                                                          |

Cloning a real clip of HAL or Rick will sound closer to the source than the built-in ffmpeg preset — the presets exist so *something* is available with zero extra install; cloning is the accurate path once you have real audio.

## End-to-end flow

```
user: "create a voice called vader from vader_clip.mp4"
|      (or Settings ▶ Runtime ▶ "VOICE / TTS" ▶ drop-zone)
▼
router_enhanced.py (voice.create pattern, ~line 1832)
  regex: (voice|clone) + (creat|make|build|clone|generat|record)
        + a media extension OR "from/using/with ... recording/audio/clip/sample"
  extracts: name ("called X") + file (a .wav/.mp3/.mp4/.m4a/.ogg/.flac/.aac/.webm path)
  → {action: CREATE_VOICE, args: {name, file}}
▼
executor_enhanced.py CREATE_VOICE handler (~line 10575)
  resolves ~ in the path, checks the file exists
  → tts_xtts.add_clone(name, path)
▼
tts_xtts.add_clone() (offline, no TTS needed)
  ffmpeg -i <clip> -ac 1 -ar 22050 <ref.wav> (mono/22.05kHz – any container, audio-only too)
  registry write: config/voices/clones.json { name: {ref_wav, language, desc} }
  reference clip stored: models/voice_profiles/<name>.wav
  → {ok: true, id: "clone:<name>", synth_ready: xtts_available()}
▼
executor sets it active: tts_router.set_active_voice("clone:<name>")
  replies: "Created the voice '<name>' ... set it active."
          (+ a note if the neural extra isn't installed yet)

— later, any time ELI speaks —

tts_router.synthesize_wav(text, voice)
  voice startswith "clone:" → tts_xtts.synthesize_wav(text, name)
  _patch_transformers_compat() (see "Defects found" below)
  os.environ["COQUI_TOS_AGREED"] = "1" (see "Defects found" below)
  _get_model() lazy-loads XTTS-v2 (~1.8GB, one-time download, cuda if free VRAM else cpu)
  model.tts_to_file(text, speaker_wav=ref.wav, language, file_path=tmp.wav)
  → WAV bytes
  None returned at ANY step → caller falls back to the normal active voice
  (never goes silent; CREATE_VOICE's own confirmation message says so up front)
```

## Files involved

| File                                        | Role                                                                                                                                                                      |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| eli/execution/router_enhanced.py (~1832)    | voice.create pattern → CREATE_VOICE                                                                                                                                       |
| eli/execution/executor_enhanced.py (~10570) | CREATE_VOICE handler → add_clone → sets active voice                                                                                                                      |
| eli/perception/tts_xtts.py                  | registry<br>(add_clone/get_clone/list_clones/delete_clone),<br>synthesis (synthesize_wav,<br>synthesize_natural_wav), availability<br>(xtts_available, natural_available) |
| eli/perception/tts_router.py                | synthesize_wav() dispatches                                                                                                                                               |
| eli/perception/voice_fx.py                  | clone:/natural:/char:/plain Piper by prefix<br>the <i>other</i> path — ffmpeg character presets,<br>unrelated dependency-wise                                             |
| config/voices/clones.json                   | clone registry (name → reference wav path,<br>language, description)                                                                                                      |

| File                             | Role                                                  |
|----------------------------------|-------------------------------------------------------|
| models/voice_profiles/<name>.wav | the normalised 22.05kHz mono reference clip per clone |

## Dependency stack

The optional extra is declared three ways in `pyproject.toml` (`clone`, `natural`, `voice` — all aliases for the same thing, worded for whichever the user searches):

```
clone    = ["torch>=2.0", "torchaudio", "torchcodec", "coqui-tts>=0.24"]
natural  = ["torch>=2.0", "torchaudio", "torchcodec", "coqui-tts>=0.24"]
voice    = ["torch>=2.0", "torchaudio", "torchcodec", "coqui-tts>=0.24"]
```

The torch trio is declared explicitly: `coqui-tts` does **not** declare `torch/torchaudio/torchcodec` itself, but imports all three at load time.

Installed from the repo root as `pip install -e ".[natural]"` (**not** `pip install "eli-v2.0[natural]"` — that string only resolves once the project itself is installed; it isn't published to PyPI under that name). On a box that already has torch+CUDA for the main model, only the delta installs.

## In the shipped AppImage (from v2.1.65)

The `natural` extra is now bundled into the **Linux AppImage** — before that it was in neither `REQUIRED_EXTRAS` nor `OPTIONAL_EXTRAS`, so every shipped build registered cloned voices and then silently spoke in Piper. Three constraints shape how it is wired:

- **CPU torch only.** `release.yml` installs `torch torchaudio` from the pytorch CPU index *before* the extras loop; default PyPI `torch` drags ~6 GB of `nvidia-*` CUDA wheels in. A guard fails the build if any `nvidia-` package appears afterwards.
- **Linux only.** `OPTIONAL_EXTRAS` is shared by all three build jobs, so `natural` is a **job-level override on build-linux** — the only job that installs the CPU stack first. Adding it globally makes windows/macOS resolve `torch` from PyPI and blows GitHub's 2 GiB per-asset limit, which fails the whole release.
- **Weights are not bundled.** The ~1.8 GB XTTS-v2 model downloads on first use. ELI is offline-by-default, so that first synthesis needs network access.

When the engine is absent the voice still registers, but synthesis falls back to Piper — and that fallback is now **visible** rather than silent: `tts_router` records what was requested and why it failed, logs it once per change, and the Settings ► Audio panel shows a NOT IN USE line naming the voice that is actually speaking. See `neural_fallback_state()`.

## Defects found and fixed while verifying this end-to-end (2026-07-28)

Installing the extra and actually cloning a voice surfaced four real problems — none of them hypothetical, all reproduced and fixed on this machine:

1. **Non-editable install shadowed the source tree.** `pip install ".[natural]"` (no `-e`) builds and installs a frozen wheel copy of `eli-v2.0` into `site-packages`, with its own `eli/eli-gui/etc.` console scripts — silently diverging from the live checkout on every future edit. Fixed by uninstalling it and reinstalling with `-e`, matching what `install.sh` always does.
2. **transformers 5.x removed a function coqui-tts still imports.** `TTS.tts.layers.tortoise.autoregressive` imports `transformers.pytorch_utils.isin_mps_friendly`, which existed only as an old Apple-MPS `torch.isin()` workaround and is gone in `transformers>=5` — the exact version this project's own `requirements.lock.txt` pins for the rest of the stack (`transformers==5.8.1`), so downgrading `transformers` was not a safe option. Fixed with a lazy compat shim in `tts_xtts._patch_transformers_compat()`: if the name is missing, define it as plain `torch.isin()`

(identical behaviour off MPS). Called from both `_get_model()` (before loading) **and** `xtts_available()/natural_avai` — the first pass only patched the model-loading path, so the availability check itself still returned False even once cloning would actually have worked.

3. **torchaudio/torchcodec missing.** coqui-tts needs torchaudio (not auto-pulled here since a newer torch was already present) and, since torch>=2.9 moved audio I/O off the old backends, torchcodec too. Installed torchaudio==2.11.0+cu130 (matching the installed torch 2.12.0+cu130, via `--index-url https://download.pytorch.org/whl/cu130`) and `coqui-tts[codec]`.
4. **Interactive TOS prompt blocks first use with no TTY.** coqui-tts gates the *first* XTTS-v2 download behind `ModelManager.ask_tos()`, an input() y/n prompt (“I have purchased a commercial license” / “otherwise I agree to the non-commercial CPML”). ELI’s GUI has no terminal for that prompt to read from — it would hang or raise `E0FError` on literally every user’s first clone, GUI or voice command alike. Fixed in `_get_model(): os.environ.setdefault("COQUI_TOS_AGREED", "1")` before importing `TTS.api` — `ModelManager.tos_agreed()` short-circuits true on that env var, same effect as answering “y” at the prompt. This is always the non-commercial path here (a local, personal voice, never redistributed — same policy the shipped voice library already applies to restricted Piper voices).

## Verified end-to-end (this session)

Ran the real path with no shortcuts — Piper-synthesised a reference clip (so the test needs no external/copyrighted audio), registered it exactly as `CREATE_VOICE` would, then synthesised *new* text (never in the reference) through `tts_xtts.synthesize_wav` directly (not the router’s fallback-friendly wrapper, so a silent fallback to a normal voice could not masquerade as a working clone):

```
[1] Piper reference clip synthesised           - 312,738 bytes
[2] add_clone("demo_clone", ref.wav)           - {"ok": true, "synth_ready": true}
[3] xtts_available()                           - True
[4] tts_xtts.synthesize_wav(new_text, "demo_clone") - real XTTS-v2 clone, non-empty WAV
[5] tts_router.synthesize_wav(new_text, "clone:demo_clone") - same result through the full app path
```

First attempt (before fix #4) returned None at step 4 — the TOS prompt raised `E0FError`, and the *router* path (step 5) had silently produced normal Piper speech instead, which is exactly the failure mode a real user would have hit invisibly. Second attempt, with the fix in place, produced genuine cloned audio at both step 4 and step 5.

## Licensing note

XTTS-v2 is Coqui’s model under the **CPML (Coqui Public Model License)** — non-commercial personal use, which is exactly what this feature is for (a local, personal voice, never redistributed by ELI). A commercial deployment would need a paid licence from Coqui directly; that is a deployment decision, not something this pipeline enforces.

## Known limitations

- First clone/natural use on a fresh install downloads ~1.8GB — no progress UI wired into the GUI yet beyond the console `tqdm` bar; a user driving this from chat/voice sees no feedback during the download.
- `synthesize_wav/synthesize_natural_wav` reload the model into the *same* process’s GPU memory as the main GGUF model when both are active — no VRAM budget coordination between the two the way vision hot-swap has (`gguf_inference._LLM_CALL_LOCK`). Fine on this 24GB-class card; worth revisiting for 8GB targets if natural/clone voices see real usage there.
- No per-clone quality/duration validation on the reference clip — a 1-second or very noisy clip is accepted and will simply clone poorly rather than being rejected up front.